

(12) **United States Plant Patent**
Hancock

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(54) **BLUEBERRY PLANT DENOMINATED**
‘KATAHDIN’

CPC A01H 5/08; A01H 6/36
See application file for complete search history.

(50) Latin Name: *Vaccinium corymbosum*
Varietal Denomination: **Katahdin**

(56) **References Cited**

U.S. PATENT DOCUMENTS

PP15,103 P3 8/2004 Hancock
PP26,031 P3 11/2015 Hancock

(71) Applicant: **Board of Trustees of Michigan State**
University, East Lansing, MI (US)

Primary Examiner — Annette H Para

(72) Inventor: **James F. Hancock**, East Lansing, MI
(US)

(74) *Attorney, Agent, or Firm* — McKee, Voorhees &
Sease, PLC

(73) Assignee: **Board of Trustess of Michigan State**
University, East Lansing, MI (US)

(57) **ABSTRACT**

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

Katandin is a new blueberry variety of *Vaccinium corym-*
bosum from the Michigan State University breeding pro-
gram. Is comprised of primarily *Vaccinium corymbosum*,
with about 7% of its genes coming from *V. darrowii*, 4%
from *V. angustifolium* and <1% from *V. tenellum* and *V.*
ashei. It is a highly productive cultivar with excellent fresh
fruit quality. It is well adapted to all northern highbush
production areas. Canes are numerous, moderately branched
and the fruit are well exposed. Its berries are large, have
small, dry picking scars, medium blue color, good firmness
and flavor. Katandin is characterized by high yields, high
vigor and strong winter hardiness.

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(22) Filed: **May 14, 2019**

(51) **Int. Cl.**
A01H 5/08 (2018.01)
A01H 6/36 (2018.01)

(52) **U.S. Cl.**
USPC **Plt./157**

(58) **Field of Classification Search**
USPC **Plt./157**

4 Drawing Sheets

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Botanical designation: The present invention relates to a
new Northern Highbush blueberry variety.

Latin name and variety denomination: The present inven-
tion relates to a new and distinct variety of *Vaccinium*
corymbosum, which is hereby denominated “Katandin.” 5

SUMMARY OF THE INVENTION

This invention relates to a new and distinct variety of
highbush blueberry plant. It has as its maternal parent MSU
46 and its Paternal parent is Draper. ‘Katandin’ is primarily
Vaccinium corymbosum, with about 7% of its genes coming
from *V. darrowii*, 4% from *V. angustifolium* and <1% from
V. tenellum and *V. ashei*. It is a highly productive cultivar
with excellent fresh fruit quality. It is likely well adapted to
all northern highbush production areas. Plants of ‘Katandin’
are vigorous and upright, although the canes can be lax when
loaded with fruit. Canes are numerous, moderately branched
and the fruit are well exposed. Its berries are large, have
small, dry picking scars, medium blue color, good firmness
and flavor.

The fruiting season of ‘Katandin’ significantly overlaps
that of the widely planted, mid-season cultivars ‘Draper’,
and ‘Bluecrop’. ‘Katandin’ has larger, firmer fruit than
‘Bluecrop’ and a much smaller picking scar. Its fruit can be
a little smaller than ‘Draper’ but otherwise has comparable
fruit quality. It is a much more vigorous than ‘Draper’ and
is more productive.

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‘Katandin’ is meant as an alternative with improved
hardiness for cold climates like Michigan. ‘Katandin’ has
high yields, high vigor and strong winter hardiness.

PRIOR VARIETIES

Among the existing varieties of blueberries, which are
known to us, and mentioned herein, ‘Draper’ U.S. Plant Pat.
No. 15,103; ‘MSU 46’ (not patented); ‘Bluecrop’ (not pat-
ented); and ‘Osorno’ (U.S. Plant Pat. No. 26,031).

ORIGIN OF THE VARIETY

Breeding procedures: Emasculated flowers of ‘MSU 46’
were pollinated in 2004 with pollen from ‘Draper’. MSU 46
is a sister of ‘Huron’ released from MSU in 2011. The seeds
were germinated, grown in a greenhouse for 1 year and then
field planted at the Southwest Michigan Research and
Extension Center (SWMREC) in Benton Harbor, Mich.
MSU 96 was first selected from a group of 121 siblings in
2009 by Jim Hancock (MSU).

ASEXUAL REPRODUCTION OF VARIETY

The original selection of MSU 96 was evaluated at the
Southwest Michigan Research and Extension Center at
Benton Harbor, Mich. from 2006-2010. Softwood cuttings
were also set in advanced trials at Grand Junction, Mich.
(MBG Marketing), Silverton, Oreg. (Oregon Blueberry
Company), Lowell, Oreg. (Fall Creek Nursery) and Grub-

benvorst, Netherlands (Schrijnwerkers Blueberries). Two-year-old plants were planted at 4×10' spacing in Michigan in 2011 and the other locations in 2012. The plantings in Michigan and Oregon were evaluated for three years, and the one in the Netherlands for two years. The plants were evaluated by Jim Hancock and Ed Wheeler (MBG Marketing) in Michigan, Peter Bouches and Dave Brazelton (Fall Creek Nursery), Bob Gabriel (Oregon Blueberry) and Leon Schrijnwerkers (Schrijnwerkers Blueberries).

STATEMENT OF STABILITY

Asexual propagation as described has demonstrated that the combination of traits that characterize this blueberry are fixed and remain true to type through at least two successive propagation cycles.

BRIEF DESCRIPTION OF PHOTOGRAPHS

The present invention will become more fully understood from the detailed description and the accompanying drawings, wherein:

FIG. 1 is a photographic print in full color of a 4-year-old Katandin blueberry plant with green fruit in a planted row in the foreground.

FIG. 2 is a photographic print in full color of a 4-year-old Katandin blueberry plant with blue fruit in a planted row in the foreground.

FIG. 3 is a photographic print in full color of a 4-year-old Katandin plant with flowers.

FIG. 4 is a photographic print in full color of fruit clusters of a 4-year-old Katandin blueberry plant. Most, but not all, of the fruit shown is mature.

DESCRIPTION OF THE NEW VARIETY

The following is a detailed botanical description of the new and distinct variety of blueberry, its flowers, fruit and foliage, based on observations of specimens grown at Benton Harbor Mich. and Grand Junction, Mich. over four years, and at South Haven, Mich., Lacota, Mich., Corvallis, Oreg., and Lowell, Oreg., over a two year period. The taxonomic characteristics chosen in the detailed description are standard in the practice (R E Gough, R J Hindle, and V G Shutak, "Identification of Ten Highbush Blueberry Cultivars using Morphological Characteristics," HortScience 11 (5): 5124, 1976). Color descriptions, except those given in common terms, are presented in Royal Horticultural Society Colour Chart designations. In cases where the color descriptions cited from The Royal Horticultural Society Colour Chart differ from the colors shown in the drawings, the colors cited from The Royal Horticultural Society Colour Chart should be considered accurate. Any deviation from these colors in the drawings is due to failure of the photographic process to exactly duplicate the colors of nature. In addition, fruit color designations in Table I are applicable only to mature fruit.

TABLE 1

'Katahdin' characteristics	
Plant traits	
Mature height	1.40 m
Mature width	1.05 m

TABLE 1-continued

'Katahdin' characteristics	
5	Height/width ratio 1.33
	Growth habit upright
	Annual renewal canes 2-4
	Internode length on spring shoots 1.56 cm average (range 1.4-1.7)
	Mature cane color Grayed green, 197 A
10	Mature cane length 1.07 m (range 80-140)
	Mature cane width 3.7 cm (range 26-41)
	Bark texture Rough and flakey
	New shoot color primarily green
	Foliage
15	Leaf shape obtuse
	Apex shape acute
	Base shape rounded
	Leaf length 5 cm average (range 4.4-5.0 cm)
	Leaf width 2.4 cm average (range 2.1-2.8 cm)
20	Leaf length/width ratio 2.1
	Leaf margin entire
	Leaf nectaries absent
	Pubescence none
	Color upper surface green (137A)
25	Color lower surface green (138B)
	Petiole length 1.7 cm average (range 1-2 cm)
	Petiole color light yellowish green (142C)
	Bud
	Bud shape ovate
30	Bud width 3.8 mm average (3.0-4.0 mm range)
	Bud length 55.2 mm average (5.0-6.0 mm range)
	Color reddish brown
	Blossoms
	Shape of corolla Round-urceolate
35	Calyx 5 lobed
	Style length 8-9 mm
	Color of open flower White with pink blushes
	Flower # per cluster 8 average (range 5-10)
	Pistil one per flower
40	Pistil color yellowish green (145A)
	Pistil length 8-9 mm
	Flower diameter 8-9 mm
	Flower length 10-11 mm
	Fragrance faint blueberry aroma
45	Calyx diameter 5 mm
	Sepals fused, 5 lobes
	Length 1-2 mm
	Width 1 mm
	Color top yellowish green (146C)
50	Color bottom yellowish green (146C)
	Reproductive organs
	Type berry
	Seed size 1.32 mm
	Number of seeds 12.7 average (ranges 5-32)
	Mature fruit
55	Height 1.1 cm average (range 1.0-1.2 cm)
	Width 1.6 cm average (range 1.5-1.7 cm)
	Shape oblate
	Color with bloom violet blue (98D)
	Color without bloom violet blue (103A)
60	Pedicel scar size range 1.5-2.0 mm (1.8 mm average)
	Pedicel length 7-8 mm
	Pedicel color yellowish green (144B)
	Peduncle length 11-13 mm
	Peduncle color yellowish green (138A)
65	Average weight 2.44 g

PERFORMANCE AND QUALITY DATA

In multi-location trials, 'Katandin' has proven to be an extremely upright very vigorous bush that ripens fruit in the early mid-season (Table 2). Its fruit have had good to

excellent size, color, firmness and flavor. It has also annually produced good to excellent yields.

In comparative Michigan trials, ‘Katandin’ has ripened a few days later than ‘Draper’ and ‘Bluecrop’ and before Legacy and Osomo (Table 3). ‘Katandin’ has a much better plant habit than ‘Bluecrop’, is more winter hardy and has a much smaller fruit scar. It is a little darker than ‘Bluecrop’ but otherwise its fruit quality is slightly better or comparable to ‘Bluecrop’. ‘Katandin’ is much more upright and vigorous than Draper, with comparable fruit quality.

The fruit weight of ‘Katandin’ was among the largest of the varieties sampled in Grand Junction in 2016 and 2017 (Table 4). It had among the firmest berries in 2016 and had similar firmness to all other varieties except ‘Duke’ in 2017. Its soluble solids were among the highest in both years, while its acidity was on the high but acceptable range in 2017, comparable to ‘Draper’.

‘Katandin’ is intended for all northern highbush production areas including Michigan, the eastern US and Canada, central Chile, Europe, SE Asia, China and the Pacific Northwest.

TABLE 2

Development and fruit characteristics of ‘Katahdin’ evaluated at one location in Michigan, two in Oregon and one in Denmark.						
		Fruit characteristics				
Location		Year	Habit	Season	Vigor	Yield
Michigan	Grand Junction	2014	9	4	7	8
		2015	9	2	8	7
		2016	9	4	8	7
Oregon	Lowell	2014	9	4	—	8
		2015	9	3	—	8
		2016	9	3	—	5
	Grubbenvorst	2014	7	4	—	5
		2015	9	3	—	8
		2016	9	3	—	8
Netherlands		2014	10	5	—	8
		2015	7	4	—	8
		Grand Mean	8.8	3.5	7.7	7.4
		Fruit characteristics				
Location		Size	Color	Scar	Firmness	Flavor
Michigan	Grand Junction	7	7	7	7	6
		8	6	8	7	7
		7	9	7	8	6
Oregon	Lowell	9	7	9	8	6
		6	6	7	6	7
		9	4	7	8	6
	Grubbenvorst	7+	5	7+	6	7
		7	6	9	5+	7
		7	6	9	6	7
Netherlands		6	6	6	6	—
		6	8	7	8	—
		Grand Mean	7.2	7.0	7.5	7.5

Two year old plants were set at 4 × 10’ spacing in 2012. Evaluations were made when the bushes were 30-50% ripe. For bush habit, 1 = sprawling, 5 = bushy and 9 = upright. For season, 1 = very early, 4-5 = mid-season and 9 = very late. The rating scale for vigor and fruit characteristics was 1-9, with 1-4 = inferior, 5-6 = acceptable, 7 = good, 8 = excellent and 9 = superior.

TABLE 2

Mean fruit ratings and ranges (parenthesis) of ‘Katahdin’ and 5 other cultivars at Grand Junction, Michigan in 2014, 2015 and 2016.					
Cultivar	Date First harvest	Plant habit	Winter hardiness	Plant vigor	Fruit Weight
Duke	6/30 (6/29-7/1)	5 (4-6)	6.5 (6-7)	6 (all 6s)	6 (4-8)
Bluecrop	7/4 (7/2-7/6)	4.7 (4-5)	6.5 (6-7)	6.9 (6-7)	6.5 (6-7)
Draper	7/4 (6/27-7/7)	4.7 (4-5)	8 (7-9)	5.5 (5-6)	7.6 (7-9)
Katandin	7/10 (7/9/-7/11)	9 (all 9s)	8.5 (8-9)	7 (all 7s)	7 (6-8)
Legacy	7/14 (7/14-7/15)	6.7 (5-7)	6 (5-7)	9 (all 9s)	6 (all 6s)
Cultivar	Fruit Color	Picking scar	Firmness	Flavor	Yield
Duke	6 (all 6s)	7 (all 7s)	8.5 (7-10)	6 (all 6s)	5.5 (5-6)
Bluecrop	6.5 (6-7)	7.5 (7-8)	8.0 (7-9)	6.5 (6-7)	5.5 (5-6)
Draper	8.5 (8-9)	8.6 (8-9)	8.8 (8-9)	7.2 (7-8)	6.5 (5-9)
Katandin	6 (4-8)	8.5 (8-9)	8.5 (8-9)	6.3 (6-7)	7.4 (7-8)
Legacy	6.5 (6-7)	7.0 (all 7s)	8.5 (8-9)	7.8 (7-8)	6.5 (7-8)

Two-year-old plants were set in 2013. Fruit evaluations were made when the bushes were 30-50% ripe. The rating scale for the vigor, hardiness and fruit characteristics were 1-9, with 1-4 = inferior, 5-6 = acceptable, 7 = good, 8 = excellent and 9 = superior. For bush habit, 1 = sprawling, 5 = bushy and 9 = upright.

TABLE 3

Average fruit physical and biochemical characteristics of ‘Katahdin’ compared to standard cultivars in Grand Junction Michigan (2016 & 2017).						
Cultivar	Year	Weight (g)	Soluble solids	Titrateable acidity	SS/TA	Firmness (g/mm)
Duke	2016	1.5a	11.6a	0.59a	19.7b	302b
	2017	2.1a	12.0a	0.81c	14.8ab	183b
Bluecrop	2016	1.6a	11.1a	0.80bc	13.9ab	202a
	2017	2.3ab	11.8a	0.67b	17.6b	163a
Draper	2016	2.1b	12.7c	0.89c	14.3ab	334h
	2017	2.3ab	11.9a	1.05d	12.1a	167a
Katandin	2016	2.1b	12.1b	0.91c	13.3a	302b
	2017	2.5b	12.7b	1.01d	12.6a	165a
Legacy	2016	2.0b	12.1b	0.73b	16.6b	301b
	2017	2.1a	13.3b	0.52a	25.6c	173ab

Three, five-fruit samples were evaluated when the bushes were 30-50% ripe.

What is claimed is:

1. A new and distinct highbush blueberry plant, substantially as described and illustrated herein.

* * * * *

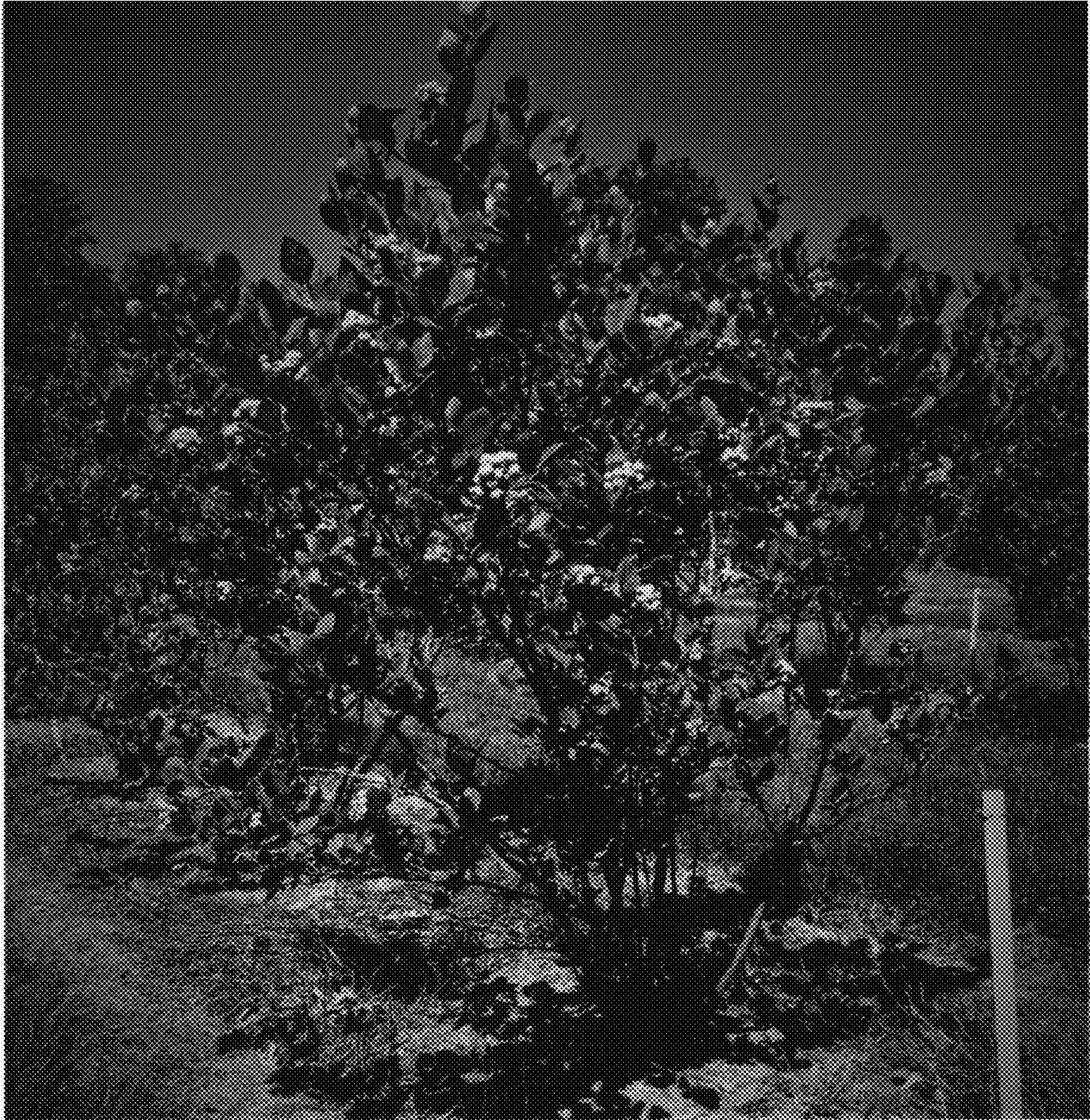


FIG. 1

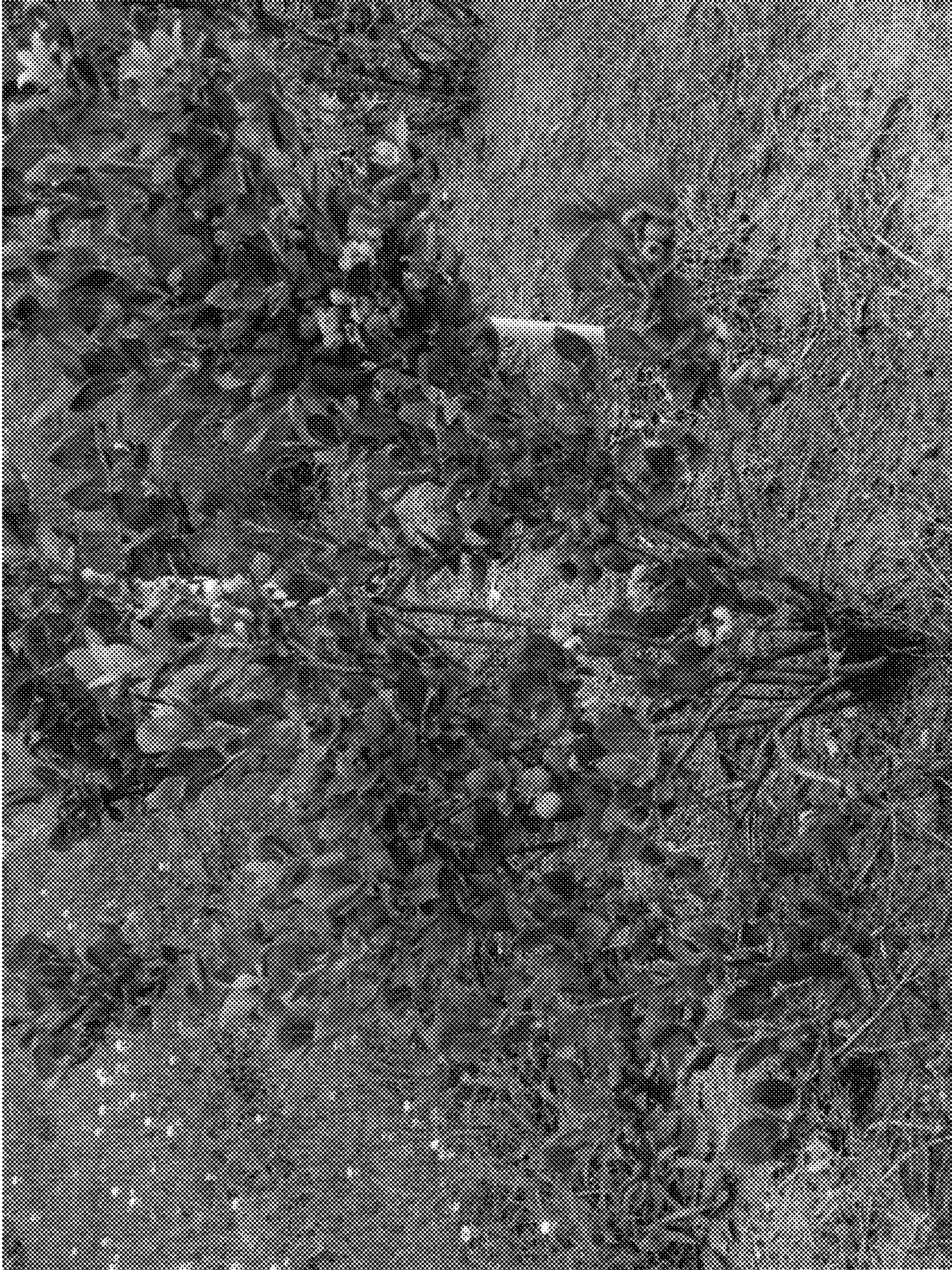


FIG. 2



FIG. 3



FIG. 4